

Date

23 August 2023

Our reference

11209669-002-HYE-0001

Number of pages

1 of 7

Contact person

Ruben White

Direct number

+31(0)88 335 8495

E-mail

ruben.white@deltares.nl

Subject

Hollandse Kust West additional morphodynamic data

Doc. Version	Author(s)	Reviewer	Approver
1.0	Ruben White Tom Roetert	Hans de Vroeg	Bas van Vossen

Background and context

The Dutch government has designated several areas in the North Sea for the development of offshore wind farms. The Hollandse Kust West Wind Farm Zone (HKW), located off the Dutch coast, is one of these areas (see Figure 1). An initial quick scan of HKW was performed by Deltares (2017), focusing on a general overview of the area, including geology, morphology and hydrodynamics. This study was further extended by Arcadis Germany GmbH and Geo-Engineering.org GmbH (2018), where geology and morphology were investigated using data sources that were available at that time and by DHI (2019), investigating metocean conditions for HKW. Finally, a detailed investigation of the morphodynamics for HKW was performed. That analysis consisted of a scour and scour mitigation study and a morphodynamic study by Deltares (2020).

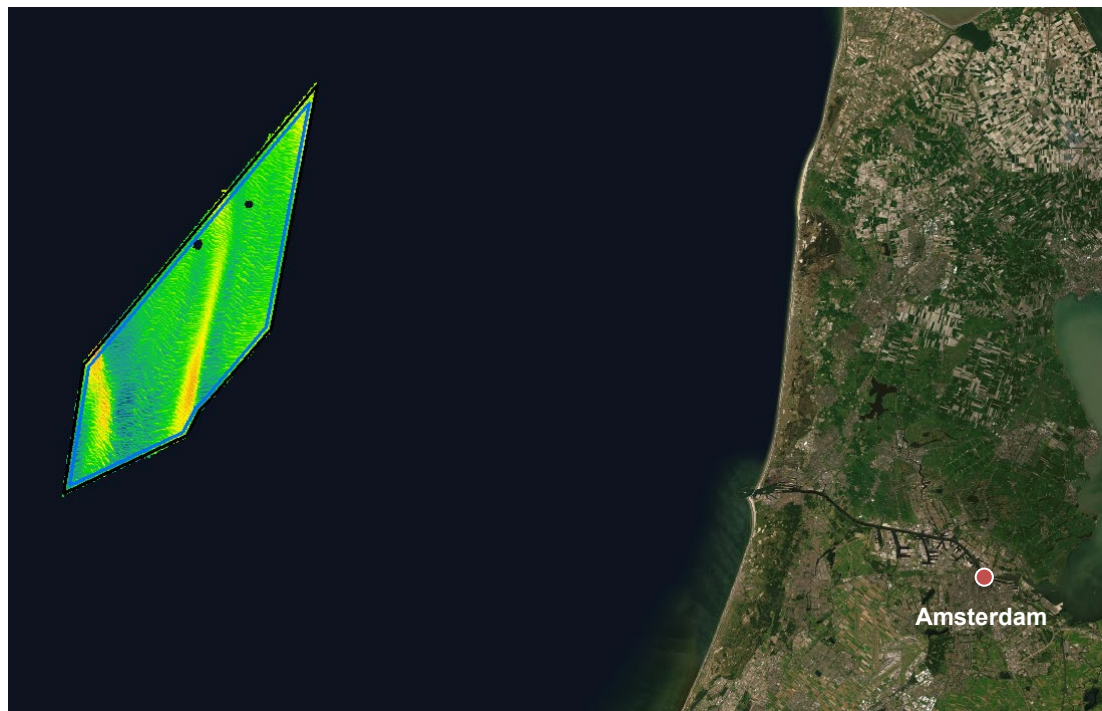


Figure 1: Location of HKW off the Dutch coast. Bathymetry data inside the windfarm extent is provided by Fugro (2019a).

Objective

The publicly available report and data for the study by Deltares (2020) only covers the period 2019 to 2059 with 5-year intervals. Deltares was requested by JBO (Jörss-Blunck-Ordemann GmbH) to use the approach applied in Deltares (2020) to predict seabed levels for the period 2025 to 2063. More specifically, the following seabed levels were requested:

- Best-Estimate prediction of the Bathymetry (BEB) for the years 2025, 2026, 2062 and 2063 (i.e., at installation and decommission dates).
- Lowest expected SeaBed Levels (LSBL) and Highest expected SeaBed Levels (HSBL) that cover the period 2025 to 2063 in 5-year intervals (i.e., the lifetime of the HKW Wind Farm).

Morphology in the Hollandse Kust West Wind Farm Area

HKW is characterised by a complex bathymetry, with a highly dynamic morphology: shore-parallel sandbanks (with wave lengths in the order of kilometres) are overlain with dynamic shore-perpendicular sand waves (wave lengths in the order of hundreds of metres), which in turn are covered by megaripples (order tens of metres wave lengths). The morphology studies by Deltares analysed the morphodynamic behaviour of the various bed forms. It was found that the sand banks can be considered stationary for the lifetime of a wind farm. Megaripples on the other hand have a high migration speed, resulting in many megaripples passing a certain location during the lifetime of a wind farm. This forms a baseline seabed level variation in the order of decimetres. The sand waves were found to have the largest impact on seabed level variations; they typically migrate fast enough (order of several meters per year) to cause metres of seabed level variation during the lifetime of a wind farm (depending on the location on the sand wave). The migration of sand waves was therefore investigated extensively in the morphology studies by Deltares.

Methodology

This memo contains a brief summary of the methodology applied in this study. An extensive description of the methodology is provided in Deltares (2020).¹

Based on the analysis of the migration speed and directions of the sand waves, predictions are made of potential seabed level variations in the period between 2025 and 2063. This resulted in a Best-Estimate prediction of Bathymetry (BEB) and an upper and lower limit of seabed levels to be expected in this period. The lower limit is named the Lowest SeaBed Level (LSBL), while the upper limit is named the Highest SeaBed Level (HSBL).

To arrive at the BEB, LSBL and HSBL a total of nine potential bathymetries are predicted for each year. These predictions are based on the statistical spreading of both migration speed and migration direction of the individual sand waves as derived from the available historical bathymetry data. These nine predictions did not result in a single bathymetry for each year, but instead generated an envelope of minimal and maximum seabed levels for each (grid) point in HKW.

The LSBL and HSBL describe the upper and lower envelope of predicted morphological seabed level changes. The LSBL₂₀₃₀ for example provides the lower envelope to be expected in the period between 2025 and 2030. Each subsequent LSBL provides the envelope between 2025 and a given year (e.g., the LSBL₂₀₃₅ provides the lower envelope to be expected in the period between 2025 and 2035). This is similar for the upper limit, which is provided by the HSBL.

¹ The methodology in Deltares (2020) describes that the bathymetry data measurements in 2018 are extrapolated to determine the future bed level predictions. In the current study this methodology is slightly adjusted for WTG 15. At this wind turbine the bathymetry data measurements in 2014-2015 were used because the measurements in 2018 do not cover this location. For all other WTG locations, the 2018 bathymetry measurements were used to determine the future bed level predictions.

Overview of the provided data

Considering the previous section it should be noted that the contents of this memo and its accompanying datafiles are the result of mathematical and/or statistical modelling and calculations, therefore the contents must be considered as Deltares’ best estimate and their accuracy is subject to usual assumptions, uncertainties and approximations made in mathematical and statistical calculations. This means that application of these values will require proper attention, including a choice for additional uncertainty bands or a safety factor. Since Deltares does not know the full project context, including possible safety factors applied in other parts of the design process, we leave it up to the Client to make a final choice for a fitting safety factor.

A table of the future bed level predictions for each of the 60 Wind Turbine Generator (WTG) locations within HKW is delivered along with this memo. The 60 WTGs are located according to the baseline layout provided by JBO (see Figure 2). The table is provided in CSV format and consists of 60 rows, one for each WTG location. The following information is given for each row: WTG ID, Easting and Northing (3 columns); BEB 2025, 2026, 2030, 2035, 2040, 2045, 2050, 2055, 2060, 2062 and 2063 (11 columns); HSBL 2025 to 2063 (11 columns); LSBL 2025 to 2063 (11 columns). A summary of this table is provided in Table 1.

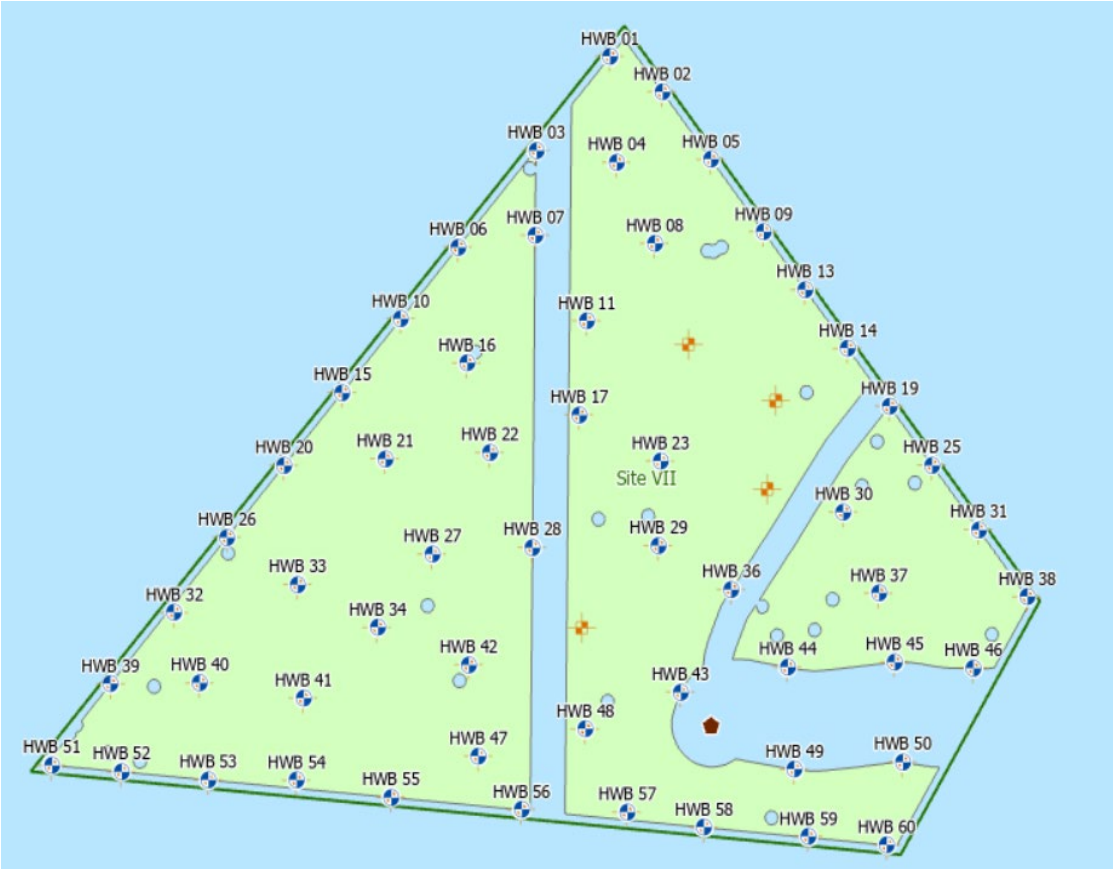


Figure 2: Baseline layout showing the locations of the 60 Wind Turbine Generators (WTG) according to JBO.

Table 1: Summary table including the Best-Estimate prediction of the Bathymetry for 2025, 2026 and 2063 [m+LAT]; and the Highest and Lowest SeaBed Levels for the period 2025-2063 [m+LAT].

WTG ID	WTG ID Prior 2023-08	BEB 2025 [m+LAT]	BEB 2026 [m+LAT]	BEB 2063 [m+LAT]	HSBL 2063 [m+LAT]	LSBL 2063 [m+LAT]
HWB 01	20	-27.97	-27.98	-28.77	-27.23	-29.49
HWB 02	15	-29.59	-29.5	-28.01	-26.56	-30.45
HWB 03	3	-28.15	-28.18	-29.22	-27.29	-30.03
HWB 04	49	-28	-28.03	-28.94	-27.36	-29.57
HWB 05	60	-29.41	-29.44	-30.36	-28.61	-30.98
HWB 06	24	-29.21	-29.24	-28	-27.13	-30.02
HWB 07	6	-28.5	-28.53	-26.24	-25.22	-29.46
HWB 08	43	-30.21	-30.22	-30.85	-29.4	-31.47
HWB 09	51	-26.72	-26.73	-26.23	-24.33	-27.47
HWB 10	42	-28.13	-28.17	-29.38	-27.27	-30.15
HWB 11	57	-28.38	-28.24	-28.39	-26.66	-29.41
HWB 12	37	-27.12	-27.12	-27.92	-26.11	-28.85
HWB 13	32	-25.68	-25.71	-26.33	-24.72	-27.05
HWB 14	47	-30.05	-29.87	-29.26	-27.26	-31.05
HWB 15	45	-28.16	-28.2	-29.31	-27.27	-30.13
HWB 16	56	-28.23	-28.27	-28.95	-27.25	-29.92
HWB 17	44	-26.54	-26.56	-27.66	-25.44	-28.54
HWB 18	28	-25.29	-25.3	-25.87	-23.71	-26.95
HWB 19	16	-30.42	-30.44	-27.26	-25.86	-31.21
HWB 20	29	-29.02	-29.04	-29.54	-28.37	-30.08
HWB 21	30	-29.21	-29.21	-29.92	-28.35	-30.59
HWB 22	14	-29	-29.02	-29.58	-27.55	-30.29
HWB 23	46	-28.75	-28.77	-27.43	-25.3	-29.87
HWB 24	18	-26.83	-26.85	-24.83	-23.54	-27.89
HWB 25	59	-30.05	-30.04	-26.23	-24.85	-30.82
HWB 26	1	-30.16	-30.17	-30.41	-29.41	-31.03
HWB 27	8	-29.26	-29.29	-27.78	-26.47	-30.32
HWB 28	52	-30.33	-30.34	-28.96	-26.93	-31.19
HWB 29	11	-27.72	-27.74	-28.64	-26.81	-29.39
HWB 30	58	-28.4	-28.41	-29.22	-27.34	-30.33
HWB 31	54	-27.58	-27.61	-28.72	-26.71	-29.71
HWB 32	9	-30.31	-30.29	-30.62	-29.38	-31.38
HWB 33	38	-30.02	-30.03	-30.17	-29.23	-30.7
HWB 34	34	-29.67	-29.71	-27.56	-26.46	-30.89
HWB 35	21	-29.17	-28.98	-27.5	-25.61	-30.09
HWB 36	40	-25.79	-25.8	-25.26	-24.24	-26.7
HWB 37	10	-27.52	-27.55	-28.54	-26.51	-29.45
HWB 38	27	-28.47	-28.48	-26.6	-24.94	-29.34
HWB 39	4	-32.42	-32.4	-30.77	-29.46	-33.11
HWB 40	31	-30.88	-30.87	-30.25	-29.23	-31.5
HWB 41	39	-27.72	-27.74	-28.74	-26.88	-29.57
HWB 42	41	-27.31	-27.35	-29.24	-25.87	-30.3
HWB 43	50	-24.37	-24.2	-25.65	-22.99	-26.51
HWB 44	12	-30.59	-30.59	-30.36	-29.42	-31.22
HWB 45	5	-31.13	-31.16	-30.86	-29.37	-32.09

HWB 46	22	-26.98	-26.99	-27.75	-26	-28.69
HWB 47	17	-29.75	-29.6	-29.66	-27.74	-30.67
HWB 48	35	-28.42	-28.23	-27.4	-24.64	-29.35
HWB 49	33	-30.19	-30.16	-26.94	-25.5	-31.04
HWB 50	25	-28.55	-28.57	-29.48	-27.53	-30.15
HWB 51	23	-31.49	-31.49	-31.58	-30.74	-32.25
HWB 52	55	-32.47	-32.48	-32.59	-31.65	-33.18
HWB 53	53	-30.8	-30.83	-31.52	-29.99	-32.21
HWB 54	7	-30.69	-30.71	-31.04	-29.98	-31.65
HWB 55	48	-29.25	-29.27	-30.02	-28.61	-30.65
HWB 56	2	-27.48	-27.29	-26.54	-24.47	-28.48
HWB 57	13	-25.52	-25.21	-26.28	-23.34	-27.26
HWB 58	26	-28.96	-28.96	-28.03	-25.96	-29.73
HWB 59	19	-30.76	-30.78	-29.02	-27.86	-31.63
HWB 60	36	-29.3	-29.29	-26.16	-24.63	-30.2

Figures of the future seabed levels are also provided along with this memo. An example of a figure showing the future bed level predictions for HWB 56 (i.e., WTG 2) in HKW is given in Figure 3. Note the shift in the cumulative seabed envelopes that occurs around 2025. This is because this study focusses on the cumulative seabed levels from 2025 onwards, while the scope of Deltares (2020) was from 2018 onwards. The period between 2018 and 2025 is shown as this clearly illustrates the evolution of the seabed levels over time. However, the cumulative envelopes are reset in 2025 to match the scope of the current study. Datafiles of the future seabed levels predictions are also provided. These datafiles are in ASCII format and provide three columns. These columns contain respectively Easting, Northing and a z-level.

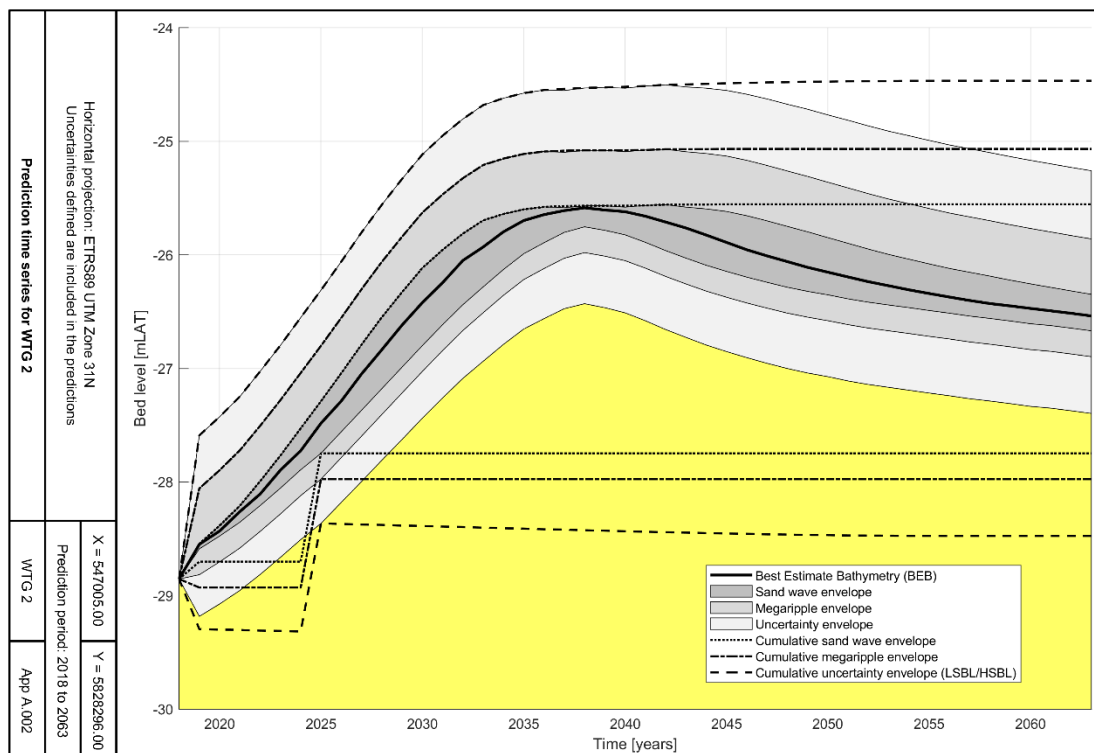


Figure 3: An example of the results for the future bed level predictions for HWB 56 (i.e., WTG 2) in HKW.

Table 2 contains an overview of the datafiles delivered along with this memo. In all datafiles Easting and Northing are provided in coordinate system WGS84 / UTM Zone 31N. The z-level is always given in metres relative to the Lowest Astronomical Tide (LAT), however the parameter reported relative to this z-level depends on the datafile (i.e., lowest expected seabed for the LSBL-files, highest expected seabed for the HSBL-files and probable seabed for the BEB-files).

Table 2: Contents of the datafiles accompanying this memo.

Datafile	Description
Future bed levels for WTGs.csv	Table with future seabed levels at WTGs
Future bed levels for WTG ##.png*	Figure with future seabed levels for WTG ##*
BEB_2025.asc	Best-Estimate prediction of Bathymetry for 2025
BEB_2026.asc	Best-Estimate prediction of Bathymetry for 2026
BEB_2030.asc	Best-Estimate prediction of Bathymetry for 2030
BEB_2035.asc	Best-Estimate prediction of Bathymetry for 2035
BEB_2040.asc	Best-Estimate prediction of Bathymetry for 2040
BEB_2045.asc	Best-Estimate prediction of Bathymetry for 2045
BEB_2050.asc	Best-Estimate prediction of Bathymetry for 2050
BEB_2055.asc	Best-Estimate prediction of Bathymetry for 2055
BEB_2060.asc	Best-Estimate prediction of Bathymetry for 2060
BEB_2062.asc	Best-Estimate prediction of Bathymetry for 2062
BEB_2063.asc	Best-Estimate prediction of Bathymetry for 2063
HSBL_2025.asc	Highest SeaBed Level for period 2025 – 2025
HSBL_2026.asc	Highest SeaBed Level for period 2025 – 2026
HSBL_2030.asc	Highest SeaBed Level for period 2025 – 2030
HSBL_2035.asc	Highest SeaBed Level for period 2025 – 2035
HSBL_2040.asc	Highest SeaBed Level for period 2025 – 2040
HSBL_2045.asc	Highest SeaBed Level for period 2025 – 2045
HSBL_2050.asc	Highest SeaBed Level for period 2025 – 2050
HSBL_2055.asc	Highest SeaBed Level for period 2025 – 2055
HSBL_2060.asc	Highest SeaBed Level for period 2025 – 2060
HSBL_2062.asc	Highest SeaBed Level for period 2025 – 2062
HSBL_2063.asc	Highest SeaBed Level for period 2025 – 2063
LSBL_2025.asc	Lowest SeaBed Level for period 2025 – 2025
LSBL_2026.asc	Lowest SeaBed Level for period 2025 – 2026
LSBL_2030.asc	Lowest SeaBed Level for period 2025 – 2030
LSBL_2035.asc	Lowest SeaBed Level for period 2025 – 2035
LSBL_2040.asc	Lowest SeaBed Level for period 2025 – 2040
LSBL_2045.asc	Lowest SeaBed Level for period 2025 – 2045
LSBL_2050.asc	Lowest SeaBed Level for period 2025 – 2050
LSBL_2055.asc	Lowest SeaBed Level for period 2025 – 2055
LSBL_2060.asc	Lowest SeaBed Level for period 2025 – 2060
LSBL_2062.asc	Lowest SeaBed Level for period 2025 – 2062
LSBL_2063.asc	Lowest SeaBed Level for period 2025 – 2063

* Individual figures of the future seabed levels for Wind Turbine Generator (WTG) 1 to 60.

References

Arcadis Germany GmbH, & Geo-Engineering.org GmbH. (2018). *Hollandse Kust (west) Wind Farm Zone Geological Desk study*. Ref: 180017; final report, dated 28 May 2018.

Deltares. (2017). Quicksan wind zones Hollandse Kust West 2-4. Ref: 11200745; memo, dated 24 February 2017.

Deltares. (2020). *Morphodynamics for Hollandse Kust (west) Wind Farm Zone*. Ref: 11204811-002-HYE-0001; final report, dated 6 July 2020. Retrieved from:
<https://offshorewind.rvo.nl/file/view/55040638/Report+-+Morphodynamical+Investigation+-+Deltares>

DHI. (2019). *Metocean desk study and database for Dutch Wind Farm Zones Feasibility level study for IJmuiden-Ver, Ten Noorden van de Waddeneilanden & Hollandse Kust (west)*. Ref: 11822658; final report, dated: 12 March 2019. Retrieved from:
<https://offshorewind.rvo.nl/file/view/55039978/Metocean+Study+%28Feasibility+report%29>.